Universidad Nacional Autónoma de México
Facultad de Ingeniería

Laboratorio de Diseño Digital VLSI

Proyecto final: sintetizador musical

Primer avance

Tarjeta EP2C5T144 (Cyclone II EP2C5T144C8)

Repositorio: https://github.com/samisal123/robot-seguidor-luz-vlsi

10 de noviembre de 2026

Figura 1. Diagrama general de bloques

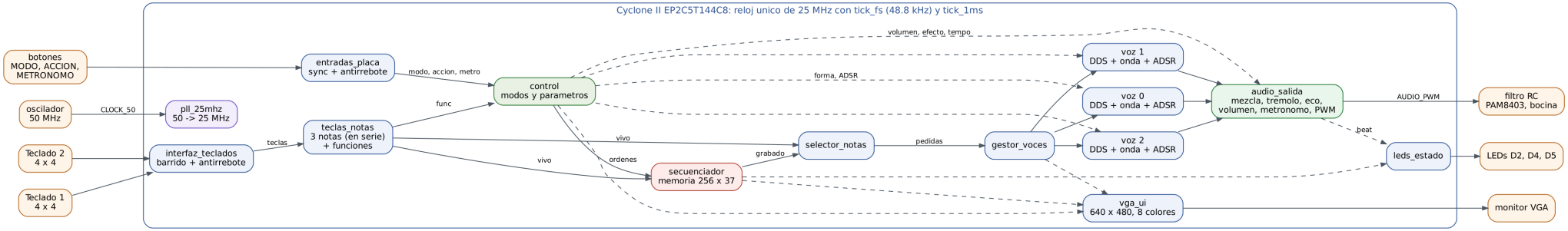


Figura 2. Proyecto en Quartus II 13.0 SP1 para la EP2C5T144C8

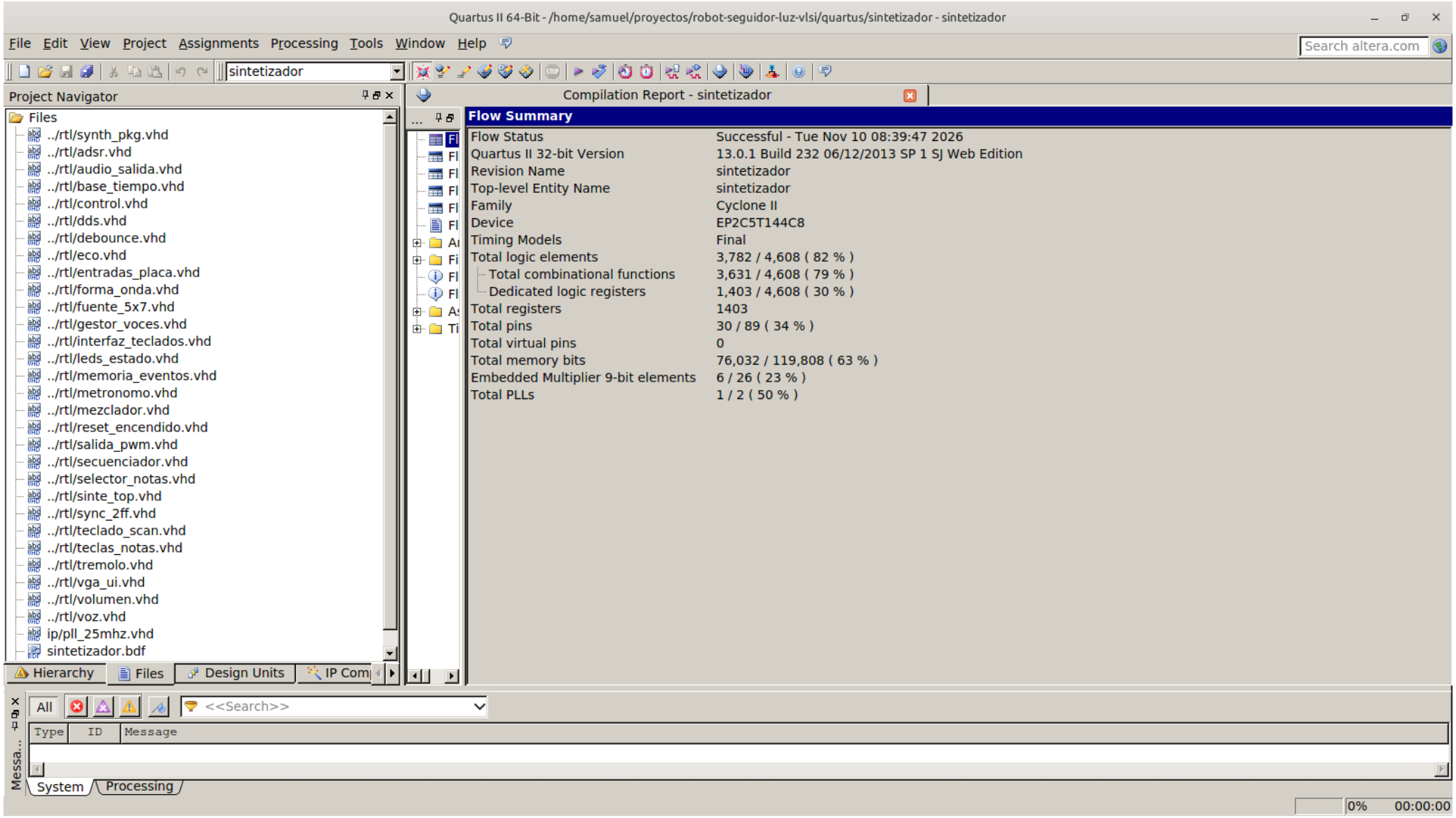


Figura 3. Creación del símbolo a partir del archivo .vhd

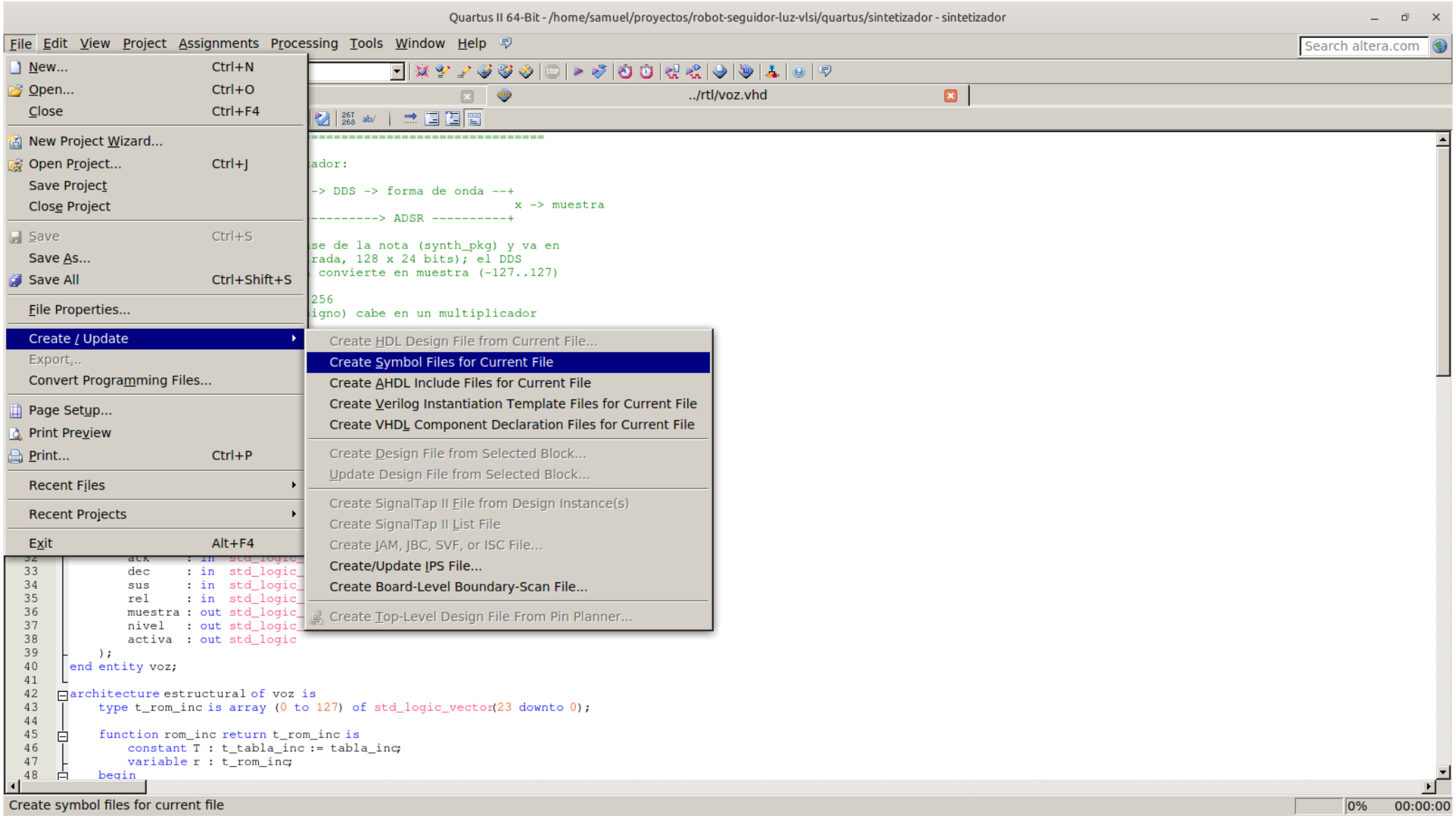


Figura 4. Diagrama de bloques del sintetizador (nivel superior)

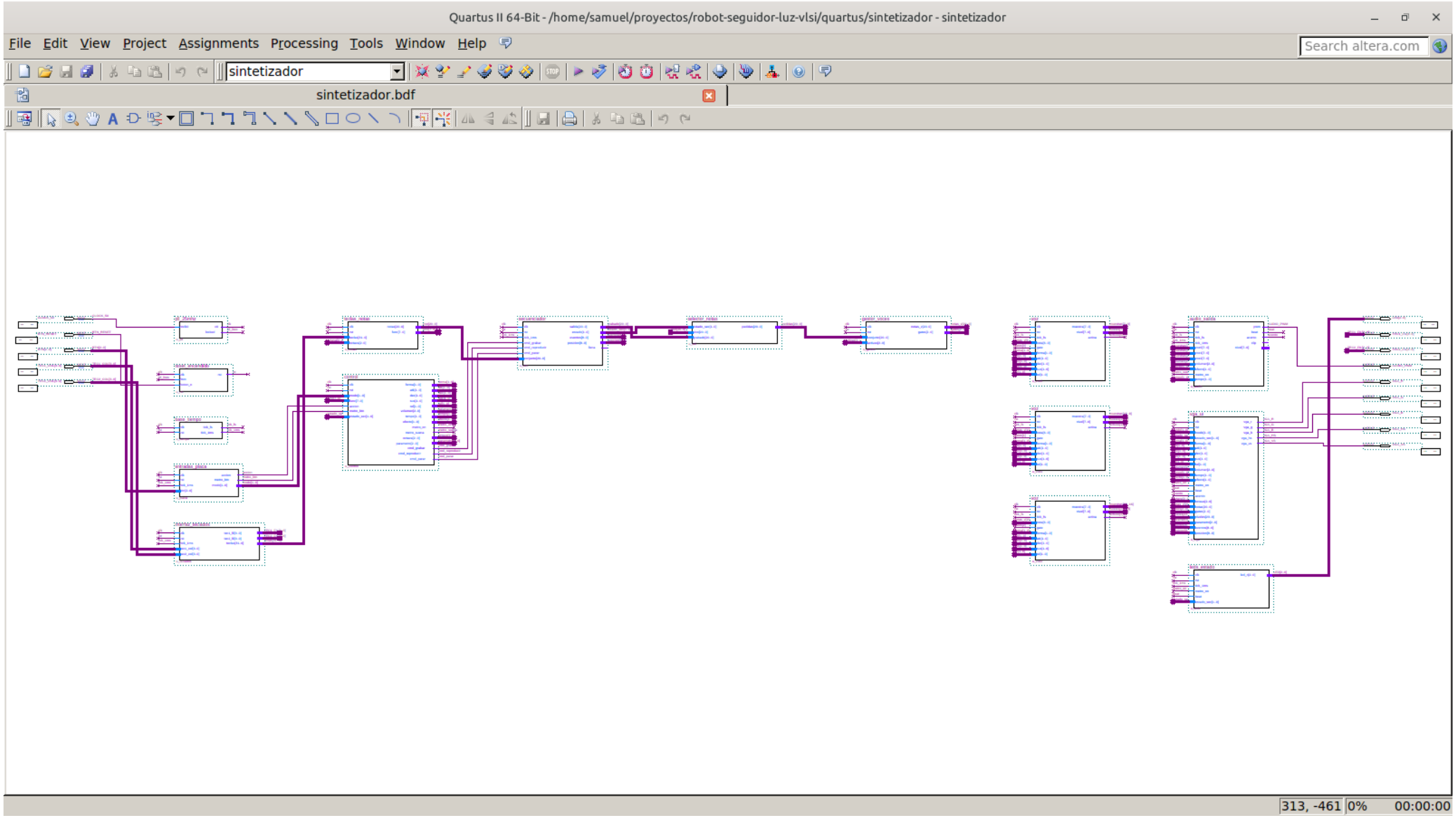


Figura 5. Diagrama de bloques (mitad derecha): gestor de voces, las 3 voces, salida de audio, VGA y LEDs

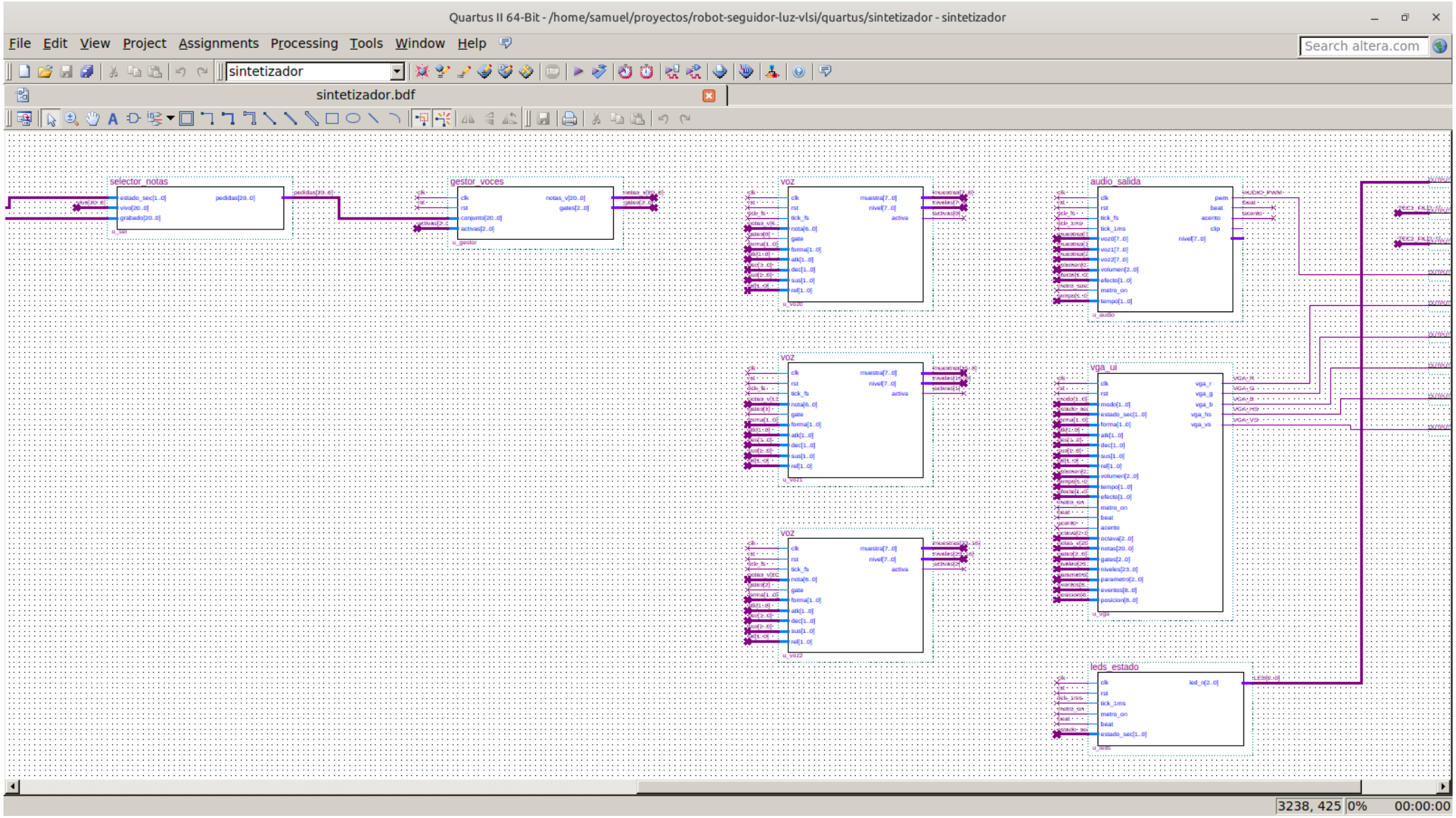


Figura 6. Diagrama de bloques (mitad izquierda): reloj, reset, botones, teclados, control, secuenciador y selector de notas

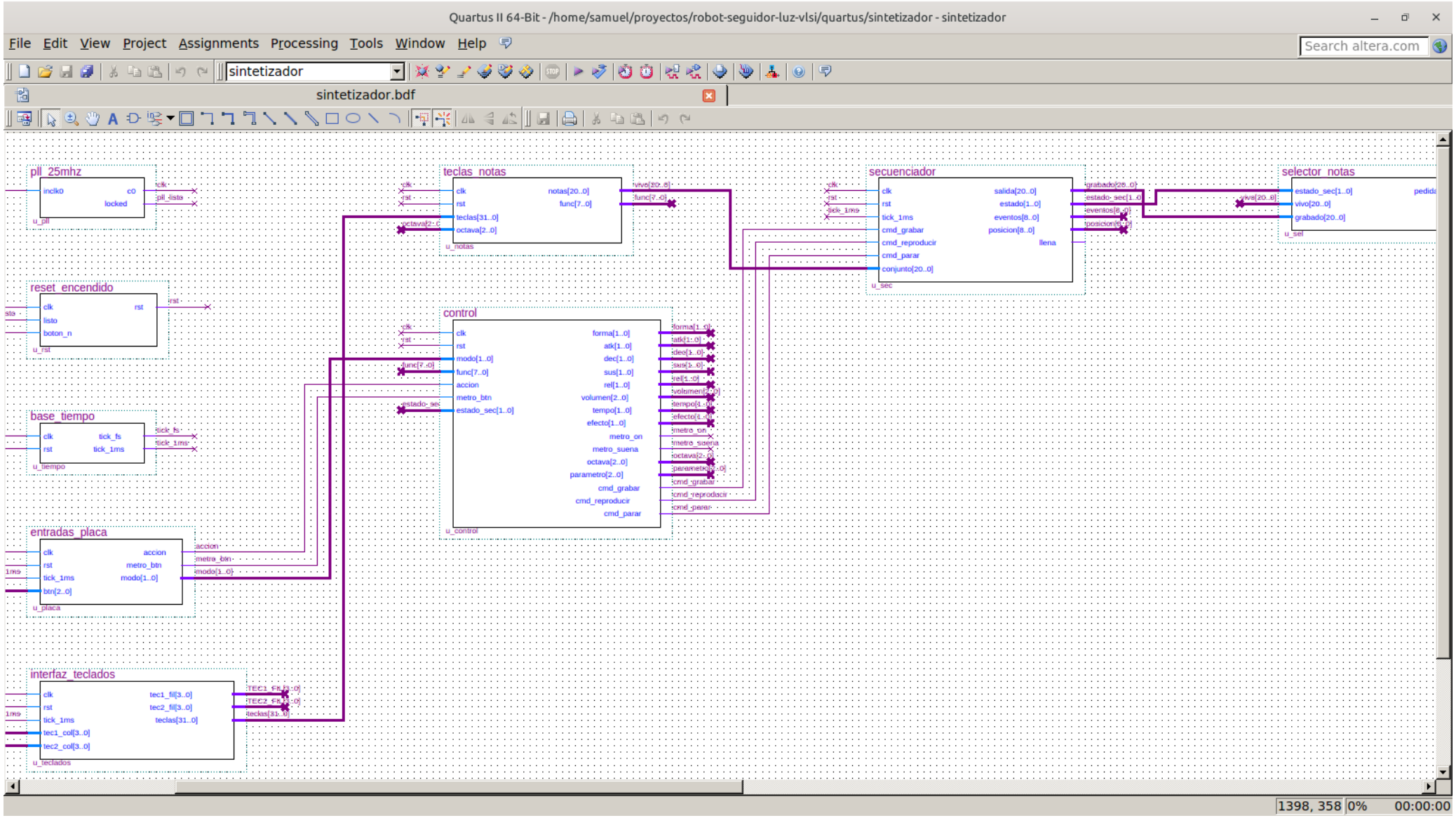


Figura 7. Jerarquía del proyecto con el diagrama de bloques como nivel superior

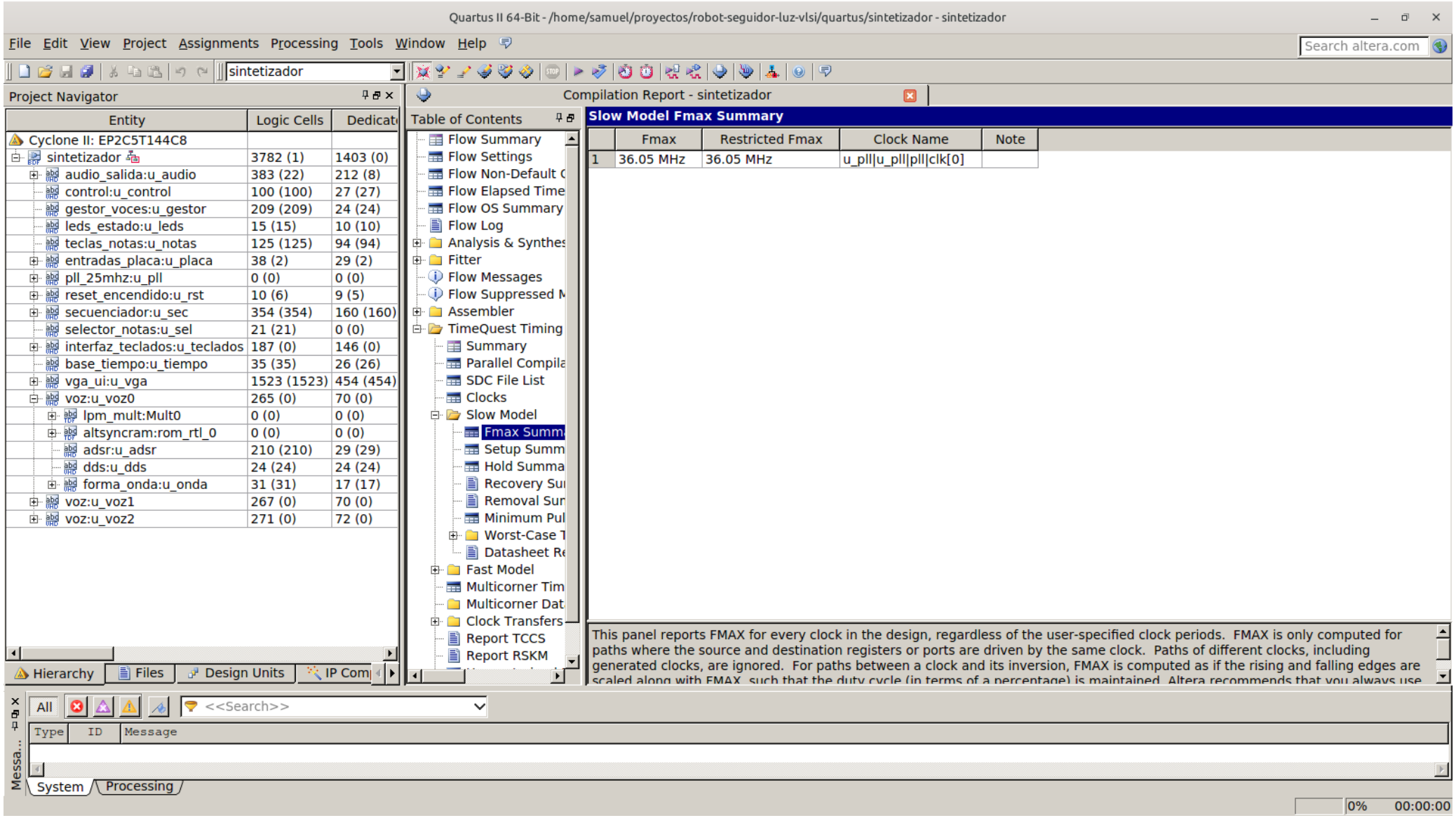


Figura 8. Resumen de la compilación

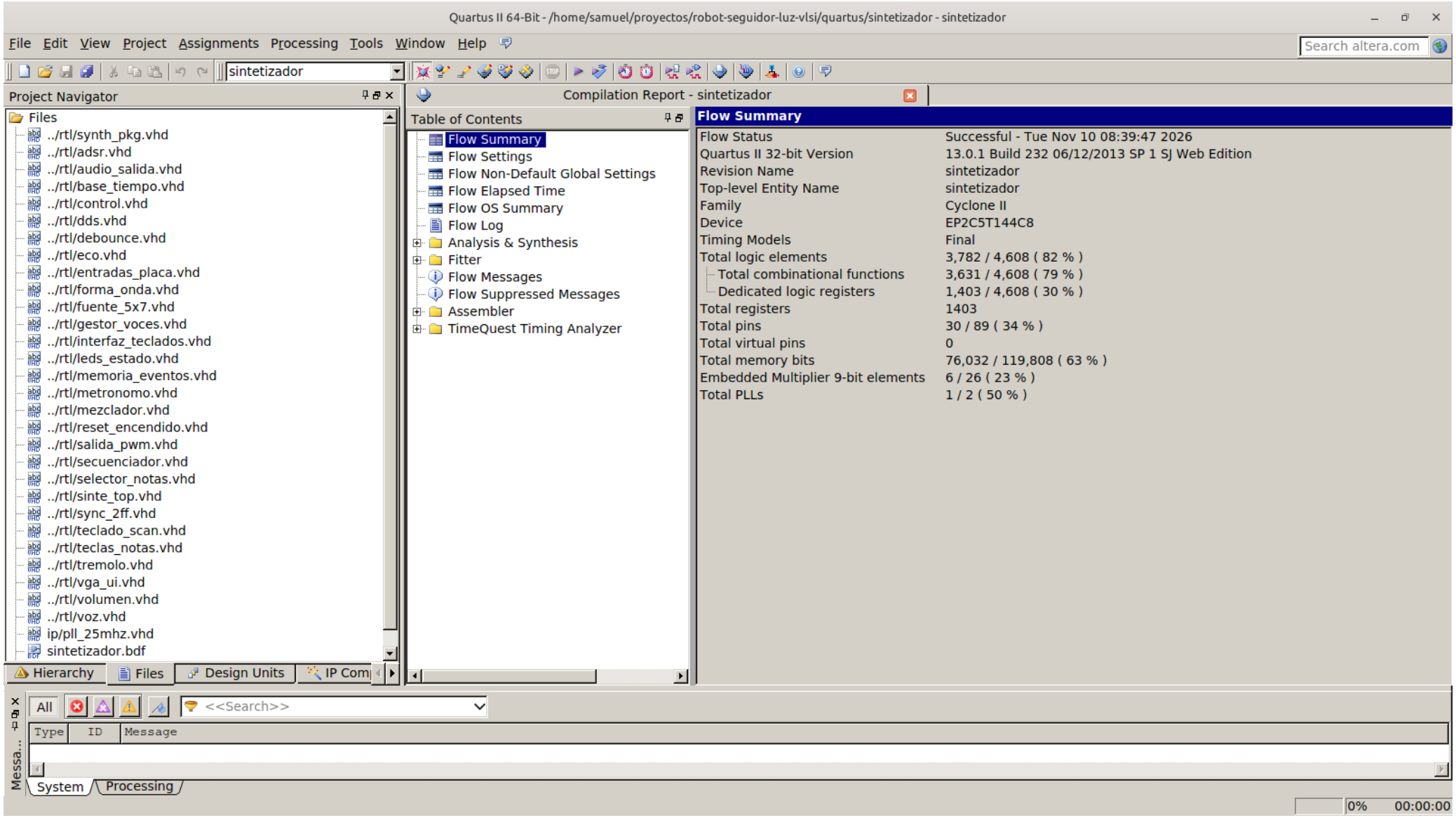


Figura 9. Recursos por módulo

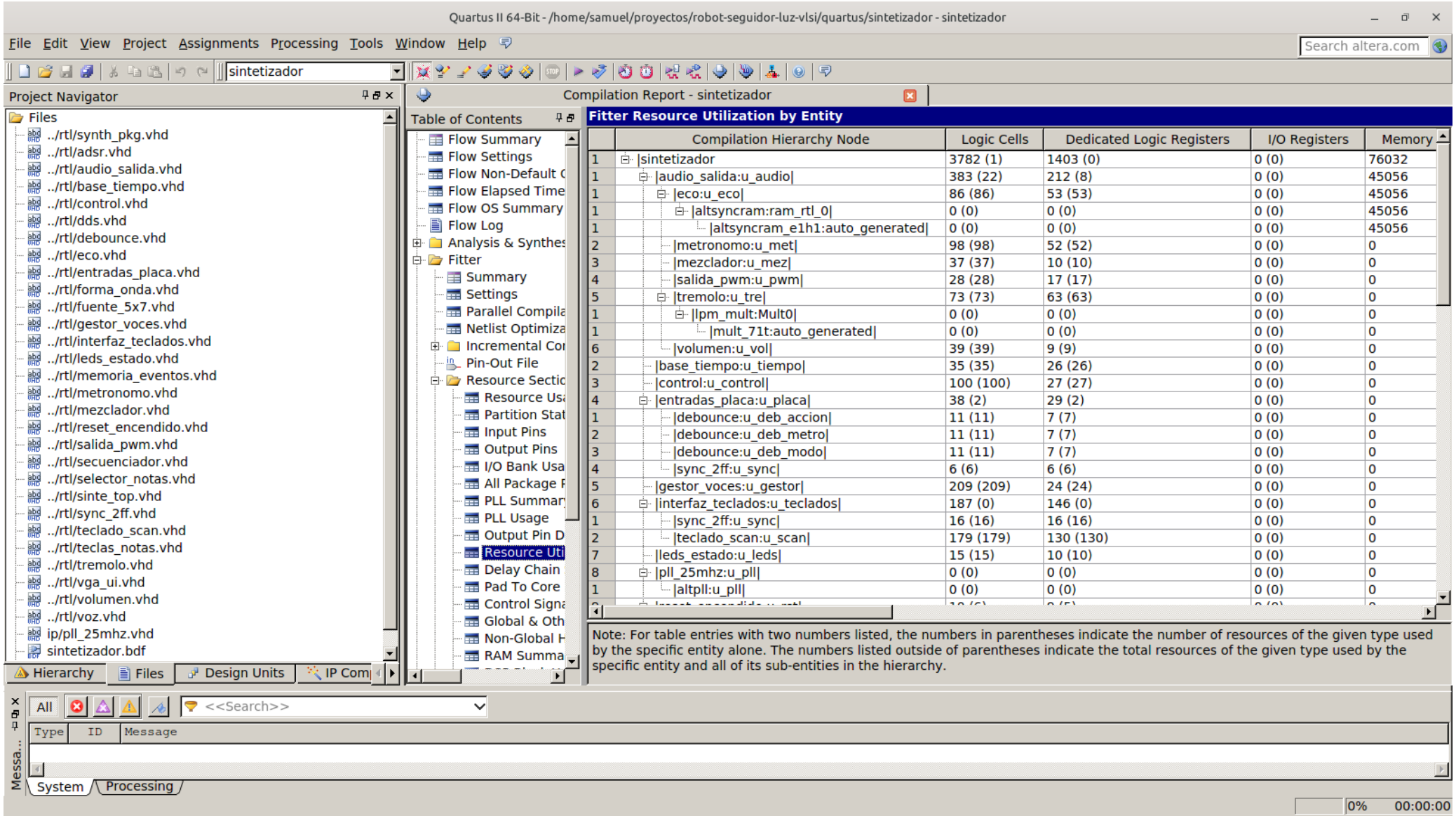
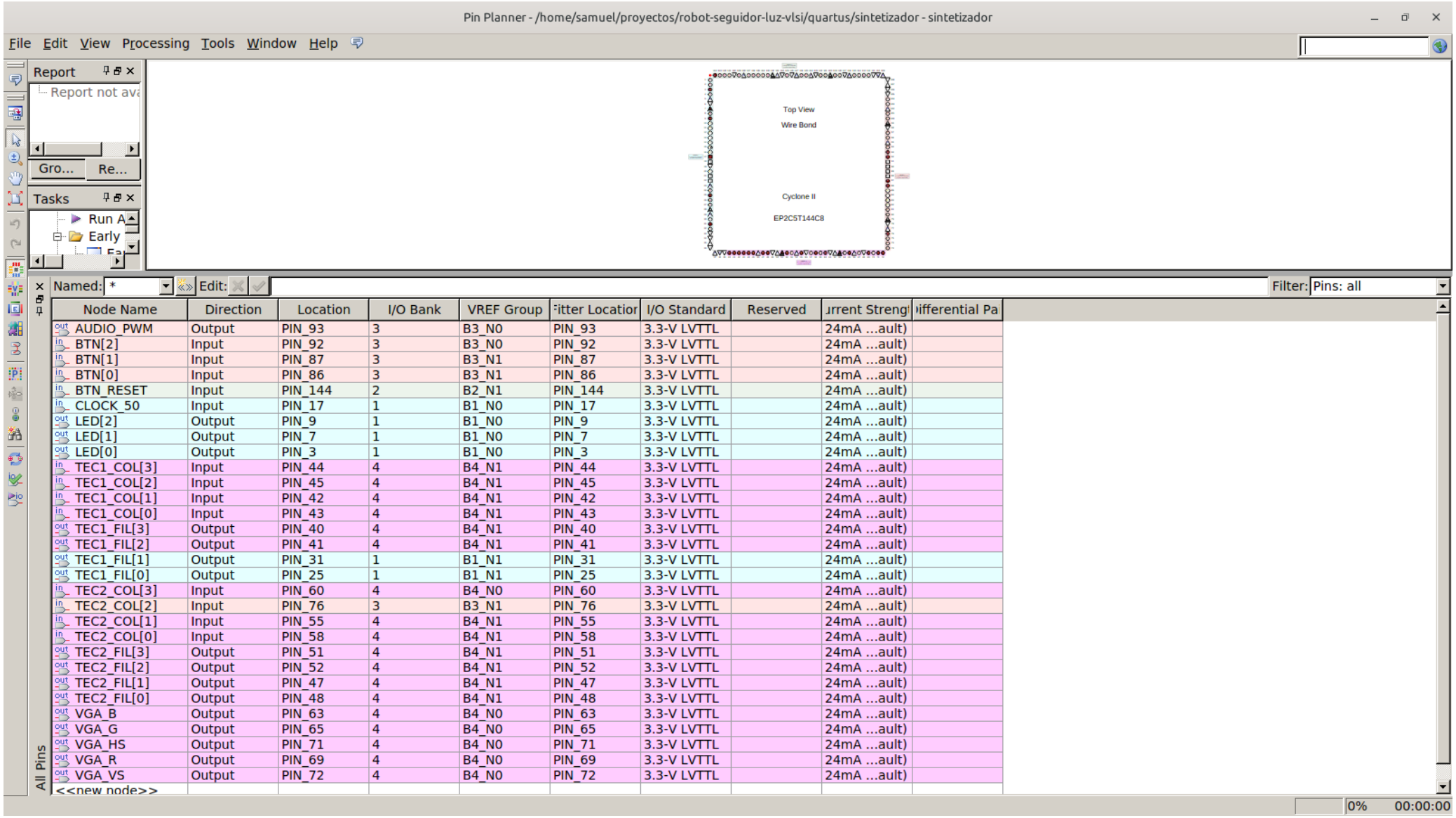


Figura 10. Asignación de pines (Pin Planner)



Node Name	Direction	Location	I/O Bank	VREF Group	itter Location	I/O Standard	Reserved	urrent Streng	ifferential Pa
AUDIO_PWM	Output	PIN_93	3	B3_N0	PIN_93	3.3-V LVTTL		24mA ...ault)	
BTN[2]	Input	PIN_92	3	B3_N0	PIN_92	3.3-V LVTTL		24mA ...ault)	
BTN[1]	Input	PIN_87	3	B3_N1	PIN_87	3.3-V LVTTL		24mA ...ault)	
BTN[0]	Input	PIN_86	3	B3_N1	PIN_86	3.3-V LVTTL		24mA ...ault)	
BTN_RESET	Input	PIN_144	2	B2_N1	PIN_144	3.3-V LVTTL		24mA ...ault)	
CLOCK_50	Input	PIN_17	1	B1_N0	PIN_17	3.3-V LVTTL		24mA ...ault)	
LED[2]	Output	PIN_9	1	B1_N0	PIN_9	3.3-V LVTTL		24mA ...ault)	
LED[1]	Output	PIN_7	1	B1_N0	PIN_7	3.3-V LVTTL		24mA ...ault)	
LED[0]	Output	PIN_3	1	B1_N0	PIN_3	3.3-V LVTTL		24mA ...ault)	
TEC1_COL[3]	Input	PIN_44	4	B4_N1	PIN_44	3.3-V LVTTL		24mA ...ault)	
TEC1_COL[2]	Input	PIN_45	4	B4_N1	PIN_45	3.3-V LVTTL		24mA ...ault)	
TEC1_COL[1]	Input	PIN_42	4	B4_N1	PIN_42	3.3-V LVTTL		24mA ...ault)	
TEC1_COL[0]	Input	PIN_43	4	B4_N1	PIN_43	3.3-V LVTTL		24mA ...ault)	
TEC1_FIL[3]	Output	PIN_40	4	B4_N1	PIN_40	3.3-V LVTTL		24mA ...ault)	
TEC1_FIL[2]	Output	PIN_41	4	B4_N1	PIN_41	3.3-V LVTTL		24mA ...ault)	
TEC1_FIL[1]	Output	PIN_31	1	B1_N1	PIN_31	3.3-V LVTTL		24mA ...ault)	
TEC1_FIL[0]	Output	PIN_25	1	B1_N1	PIN_25	3.3-V LVTTL		24mA ...ault)	
TEC2_COL[3]	Input	PIN_60	4	B4_N0	PIN_60	3.3-V LVTTL		24mA ...ault)	
TEC2_COL[2]	Input	PIN_76	3	B3_N1	PIN_76	3.3-V LVTTL		24mA ...ault)	
TEC2_COL[1]	Input	PIN_55	4	B4_N1	PIN_55	3.3-V LVTTL		24mA ...ault)	
TEC2_COL[0]	Input	PIN_58	4	B4_N1	PIN_58	3.3-V LVTTL		24mA ...ault)	
TEC2_FIL[3]	Output	PIN_51	4	B4_N1	PIN_51	3.3-V LVTTL		24mA ...ault)	
TEC2_FIL[2]	Output	PIN_52	4	B4_N1	PIN_52	3.3-V LVTTL		24mA ...ault)	
TEC2_FIL[1]	Output	PIN_47	4	B4_N1	PIN_47	3.3-V LVTTL		24mA ...ault)	
TEC2_FIL[0]	Output	PIN_48	4	B4_N1	PIN_48	3.3-V LVTTL		24mA ...ault)	
VGA_B	Output	PIN_63	4	B4_N0	PIN_63	3.3-V LVTTL		24mA ...ault)	
VGA_G	Output	PIN_65	4	B4_N0	PIN_65	3.3-V LVTTL		24mA ...ault)	
VGA_HS	Output	PIN_71	4	B4_N0	PIN_71	3.3-V LVTTL		24mA ...ault)	
VGA_R	Output	PIN_69	4	B4_N0	PIN_69	3.3-V LVTTL		24mA ...ault)	
VGA_VS	Output	PIN_72	4	B4_N0	PIN_72	3.3-V LVTTL		24mA ...ault)	
<<new node>>									

Figura 11. Frecuencia máxima (TimeQuest)

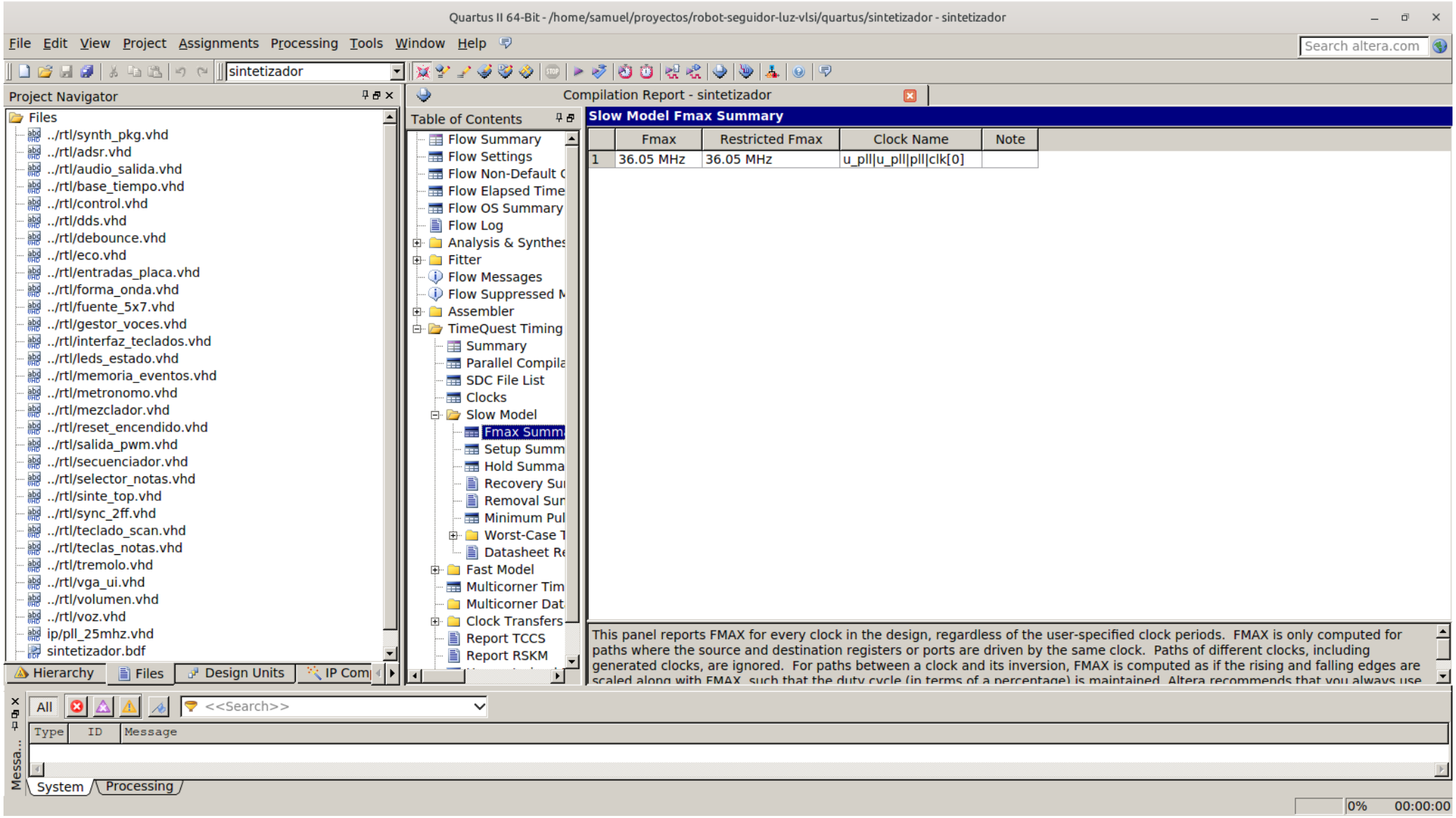


Figura 12. Vista RTL del nivel superior

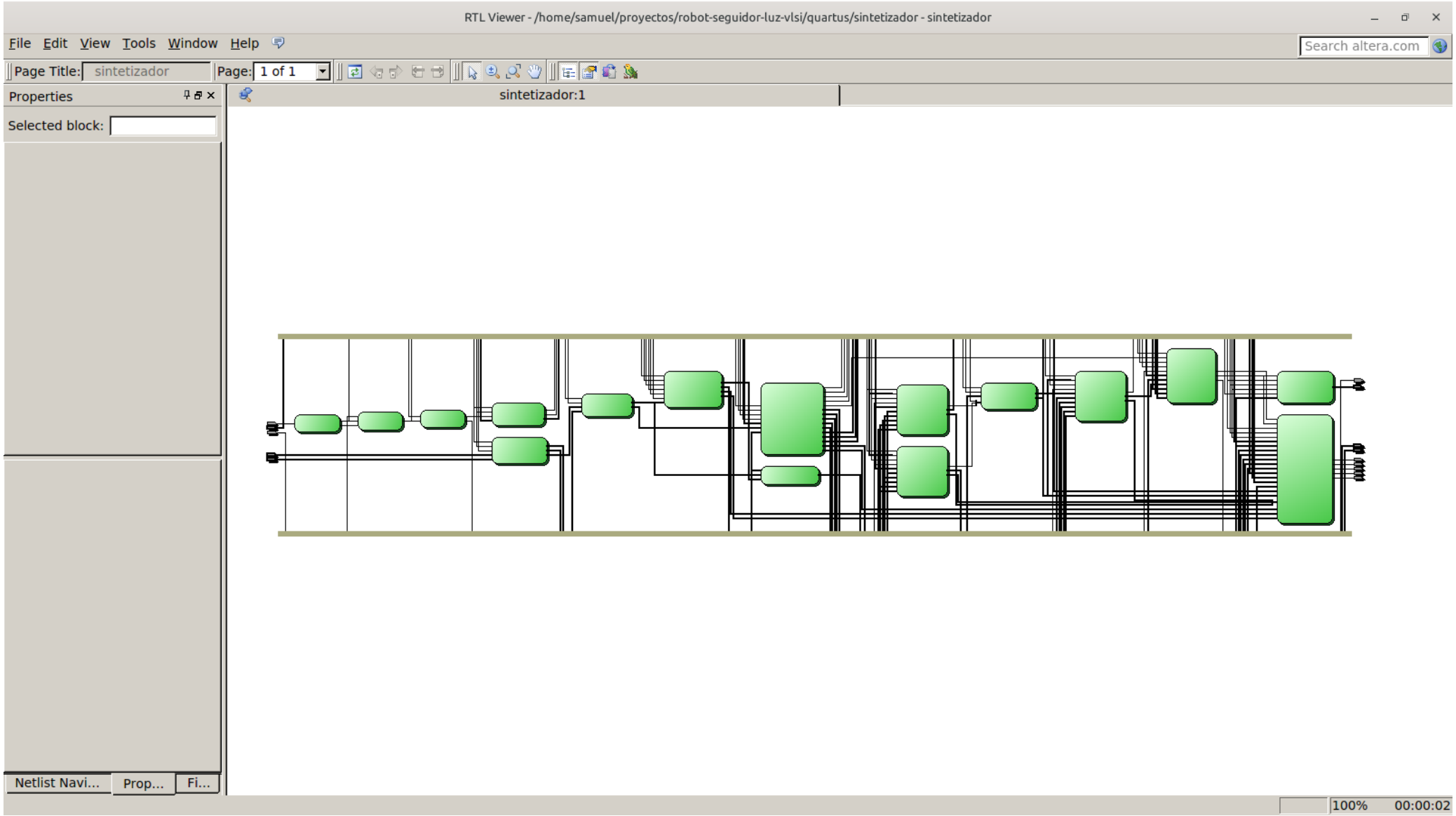
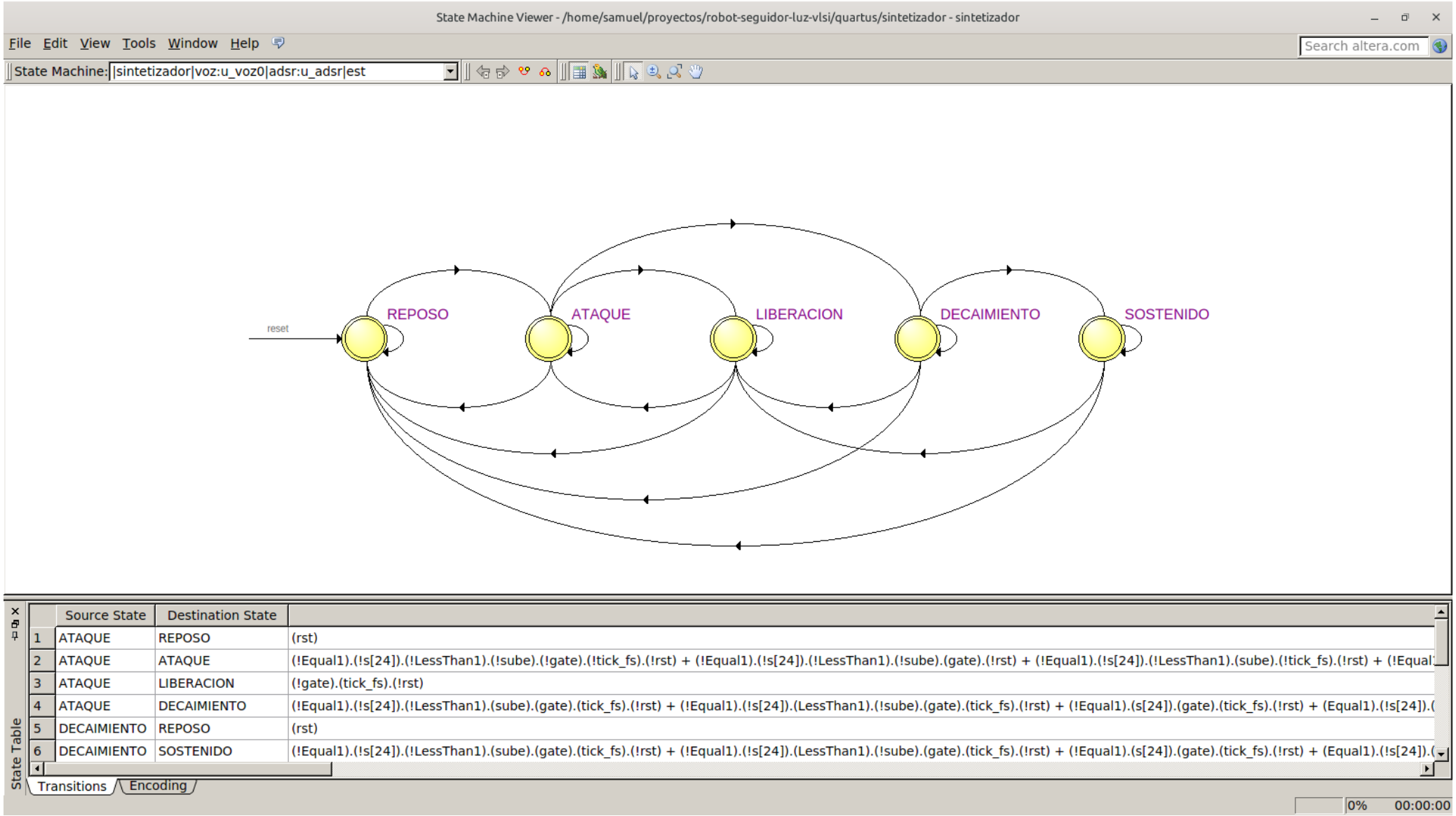


Figura 13. Máquina de estados del ADSR (State Machine Viewer)



	Source State	Destination State	
1	ATAQUE	REPOSO	(rst)
2	ATAQUE	ATAQUE	(!Equal1).(!s[24]).(!LessThan1).(!sube).(!gate).(!tick_fs).(!rst) + (!Equal1).(!s[24]).(!LessThan1).(!sube).(gate).(!rst) + (!Equal1).(!s[24]).(!LessThan1).(sube).(!tick_fs).(!rst) + (!Equal
3	ATAQUE	LIBERACION	(!gate).(tick_fs).(!rst)
4	ATAQUE	DECAIMIENTO	(!Equal1).(!s[24]).(!LessThan1).(sube).(gate).(tick_fs).(!rst) + (!Equal1).(!s[24]).(LessThan1).(!sube).(gate).(tick_fs).(!rst) + (!Equal1).(s[24]).(gate).(tick_fs).(!rst) + (Equal1).(!s[24]).(
5	DECAIMIENTO	REPOSO	(rst)
6	DECAIMIENTO	SOSTENIDO	(!Equal1).(!s[24]).(!LessThan1).(sube).(gate).(tick_fs).(!rst) + (!Equal1).(!s[24]).(LessThan1).(!sube).(gate).(tick_fs).(!rst) + (!Equal1).(s[24]).(gate).(tick_fs).(!rst) + (Equal1).(!s[24]).(

Figura 14. Símbolos de los módulos con sus puertos

